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Editorial Board
Annals of Mathematics
Princeton University & IAS

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

"The Riemann Hypothesis via Discrete Spectral Theory: Zeros of the Zeta Function as Eigenvalues of a Self-Adjoint Lattice Operator,"

for consideration for publication in *Annals of Mathematics*.

Summary. The paper pursues the Hilbert-Pólya approach to the Riemann Hypothesis: constructing an explicit self-adjoint operator whose spectrum corresponds to the non-trivial zeros of the Riemann zeta function $\zeta(s)$. We formulate the problem on a discrete spectral lattice, where self-adjointness automatically forces all eigenvalues to be real—placing all zeros on the critical line $\mathrm{Re}(s) = 1/2$.

Approach. The construction is motivated by: (i) Montgomery's pair correlation conjecture, which links zero spacings to GUE random matrix eigenvalue statistics; (ii) the work of Berry and Keating on the classical Hamiltonian $H = xp$; and (iii) the spectral geometry of arithmetic surfaces. The discrete formulation provides an explicit, finite-dimensional Hamiltonian whose adjacency spectrum converges to the zeta zeros as the lattice refines.

Suggested Reviewers.

1. **Prof. Alain Connes** (IHES/Collège de France) — Author of the spectral interpretation of the Riemann zeros via noncommutative geometry.
2. **Prof. Peter Sarnak** (Princeton/IAS) — Leading expert on L-functions, random matrix theory, and the Riemann Hypothesis.
3. **Prof. Michael Berry** (Bristol) — Developer of the Berry-Keating Hamiltonian approach to the Riemann Hypothesis.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Respectfully submitted,

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